

XIA MENG

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EDUCATION

Hainan University

Sep 2023–Jul 2027

Bachelor's Degree in Agriculture Education

Haikou, China

- **Average Score:** 87.65/100 | **Rank:** Top 7.14%
- **Core Coursework:** Genetics; Plant Physiology and Biochemistry; Microbiology; General Agricultural Machinery; Agricultural Information Technology; Agricultural Internet of Things Applications.

PUBLICATIONS

- Yi Min, Yanli Zhong, Tingting Jiang, Xiao Wei, **Xia Meng**, Deqiu Li, Peng Zeng, Meihong He, Chuntao Wei, Jianbao Wu, Yuxuan Qin, Diqi Yang, and Tingting Yu. "Differences in Oropharyngeal Viromes Between Surviving and Deceased Malayan Pangolins in a Rescue Context." *Microorganisms*, vol. 14, no. 9, Article 2043, 2026. DOI: <https://doi.org/10.3390/microorganisms14092043>

RESEARCH EXPERIENCE

X-shredding Gene Drive Strategy in the Melon Fly

Aug 2026–Present

- Contribute to sgRNA module construction across three plasmid configurations (D, E, and D+E), including plasmid-map review, primer design, PCR amplification, agarose gel electrophoresis, and gel extraction.
- Perform E. coli transformation, colony picking and expansion, colony PCR, plasmid extraction, DNA quantification, track construct verification for subsequent functional studies with Cas9.
- Extract genomic DNA from *Bactrocera cucurbitae* samples and perform embryonic microinjection on *Bactrocera cucurbitae*, preparing material for downstream molecular detection and genotyping.

Virome and Survival Outcomes in Rescued Malayan Pangolins

Sep 2025–Feb 2026

- Screened target viral and representative reference sequences and curated annotations, taxonomic information, and sample metadata for phylogenetic reconstruction.
- Aligned and quality-checked pestivirus-like, gammaretrovirus-like, and picorna-like sequences; used MAFFT, IQ-TREE, and FigTree for maximum-likelihood tree reconstruction, branch-support assessment, and annotation.
- Integrated phylogenetic placement, viral abundance, and sample distribution to compare survivors and non-survivors, distinguish potential host-associated viral signals from those of dietary or environmental origin, and support figure preparation and manuscript documentation.

PATENT APPLICATIONS

- High-throughput microbial activity and antimicrobial susceptibility detection system based on dynamic optical density tracking. Application No. 202620417949.X.
- Two-phase temperature-controlled anti-aggregation system based on composite cooling, graphene heating, and temperature-difference compensation. Application No. 202620418623.9.
- Microbial single-photon detection system integrating multispectral light sources and a nanophoton counter. Application No. 202620418884.0.

INTERNSHIP AND SUMMER SCHOOL EXPERIENCE

Shenzhen Bay Laboratory (SZBL) | Research Intern

Jul–Aug 2026

SZBL–SMART Jointly Organized Project | Supervisor: Dr. Guoxun Wang

- Contributed to a murine norovirus NS3 project, combining recombinant protein preparation with literature analysis of its roles in viral replication and egress.
- Cloned NS3 into pET-28a through PCR amplification and homologous recombination, followed by transformation into E. coli TOP10 and screening of 3–5 candidate clones for sequence verification.
- Expressed sequence-verified constructs in E. coli BL21 using IPTG induction at 16°C and purified His-tagged NS3 by affinity chromatography, followed by dialysis and low-temperature storage for subsequent characterization.

SMART International Summer School

Jul–Aug 2026

Shenzhen Medical Academy of Research and Translation | Shenzhen, China

- Explored research approaches spanning molecular mechanisms and cellular and tissue function through symposia on intracellular calcium signaling, mitochondrial molecular machinery, microbial systems biology, and tissue self-organization.
- Studied applications of cryo-electron microscopy and computational modeling to mitochondrial structure and function, alongside molecular reconstitution and quantitative cell biology approaches to tissue assembly.
- Examined the translation of biological discoveries into therapies through lectures on non-viral extracellular vesicle gene delivery, clinical development, and intellectual property strategy, connecting basic research with therapeutic delivery and development considerations.

HONORS AND AWARDS

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| • First-Class Comprehensive Scholarship, Hainan University | 2025 |
| • Second-Class Agricultural Major Scholarship, Hainan University | 2025 |
| • Third-Class Agricultural Major Scholarship, Hainan University | 2024 |
| • Third Prize , 12th Challenge Cup, Hainan Provincial Undergraduate Extracurricular Academic Science and Technology Competition | 2025 |
| • Silver Award , China International College Students' Innovation Competition, Hainan University round, Undergraduate Creative Group | 2025 |
| • Second Prize , 19th Challenge Cup, Hainan University Undergraduate Extracurricular Academic Science and Technology Competition | 2025 |
| • Silver Award , FLTRP·ETIC Understanding Contemporary China National College Students' Foreign Language Proficiency Competition, university round, Short Video category | 2024 |

TECHNICAL SKILLS

Molecular biology: Primer design, PCR, gel electrophoresis and extraction, seamless cloning, bacterial transformation, colony screening, and plasmid extraction.

Bioinformatics: Sequence screening and curation, multiple sequence alignment, phylogenetic analysis and visualization; MAFFT, IQ-TREE, FigTree.